

Lab Experiment: Virtualization in Cloud Computing

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Part 1 — Concepts: Hypervisors and Virtualization

1. Which type of hypervisor do you think AWS, Azure, and GCP run on their physical servers, and why does that matter for performance and multi-tenancy?

Cloud providers like AWS, Azure, and GCP run on **Type 1 (Bare-Metal)** hypervisors (such as customized KVM, Xen, or AWS Nitro). This matters significantly for performance because there is no underlying Host Operating System taking up overhead resources. It also provides strict, hardware-level security and isolation which is absolutely necessary for multi-tenancy (hosting multiple different customers on the same physical machine safely).

2. What is the difference between a hypervisor and an operating system?

An operating system manages physical hardware to run software applications and provide an environment for a single user. A hypervisor, on the other hand, specifically abstracts and divides physical hardware to run and manage multiple, isolated Virtual Machines, each of which contains its own separate operating system.

3. Name one advantage and one disadvantage of virtualization for a cloud provider.

Advantage: Vastly improved hardware utilization. Instead of servers sitting idle, providers can pack multiple customers onto one physical server, maximizing efficiency and lowering costs.

Disadvantage: It introduces a single point of failure. If the underlying physical server's hardware fails, all of the virtual machines hosted on that specific server will crash simultaneously.

Part 2 — Hands-On: Creating Your First Virtual Machine

Host vs. Guest Resources Comparison

Reflection: Why do the numbers differ, and who is responsible for enforcing that the guest cannot see or use more than what was allocated?

The numbers differ because the Virtual Machine is only given a restricted “slice” of the physical computer's total resources. The hypervisor (VirtualBox) is responsible for acting as the boundary and enforcing these limits so the guest OS cannot see, access, or consume the host's remaining hardware.

Resource	Host (MacBook Pro 16-in, 2019)	Guest (CloudLab-VM1)
CPU cores visible	8 Cores (Intel Core i9-9880H)	1 Core
RAM visible	16 GB (2667 MHz DDR4)	2048 MB (2 GB)
Disk size visible	1 TB	15 GB

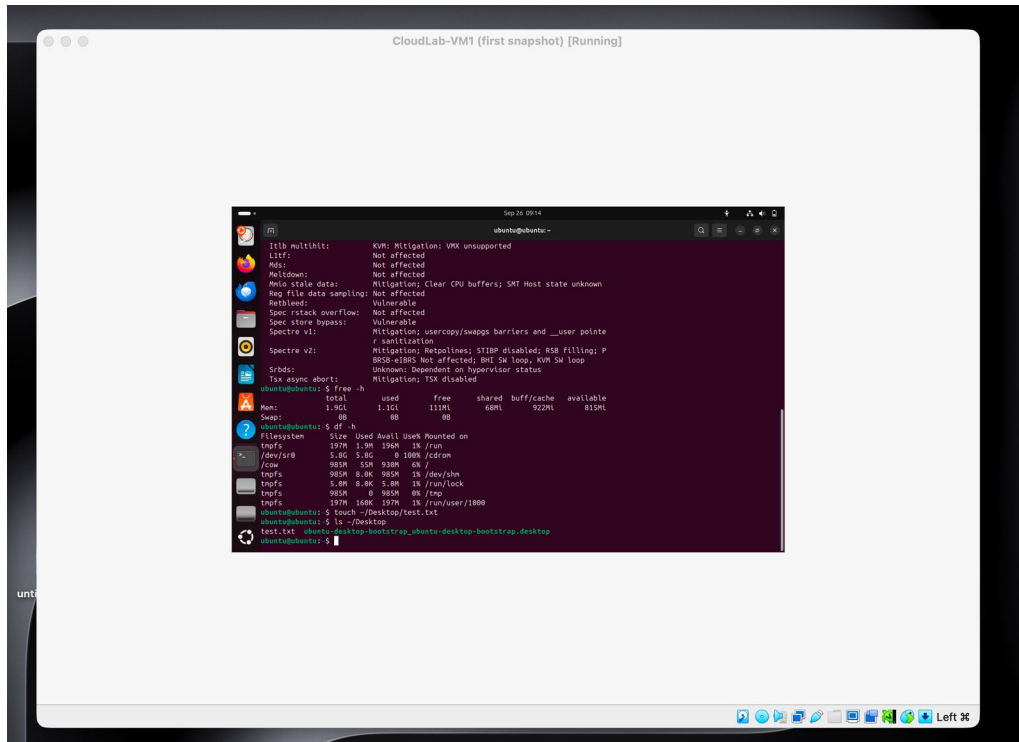


Figure 1: Terminal output inside CloudLab-VM1 showing `lscpu`, `free -h`, and `df -h` — the guest’s view of allocated CPU, RAM, and disk resources.

Part 3 — Snapshots and Cloning

Step 2 Reflection: How is this similar to how a cloud provider might let you “roll back” a server, or spin up a new instance from a saved machine image? This is exactly how cloud providers allow users to roll back a server to a previous state. They use saved block-storage snapshots or Machine Images (like AWS AMIs) to provision a fresh, pre-configured server in seconds without having to manually reinstall the operating system and dependencies from scratch. In this lab, a snapshot named `first snapshot` was taken, a test file (`test.txt`) was created on the desktop afterward, and restoring the snapshot removed that file — confirming the VM returned exactly to its saved state.

Step 3 Reflection: In a public cloud, this cloning operation happens automatically when you increase the number of instances in an auto-scaling group. What manual steps did you just do that a cloud platform automates?

I had to manually right-click the machine, choose Clone, select Full Clone, name the new instance, and manually press the start button. In a public cloud, an Auto-Scaling group automates all of this by monitoring traffic, making API calls to clone a base im-

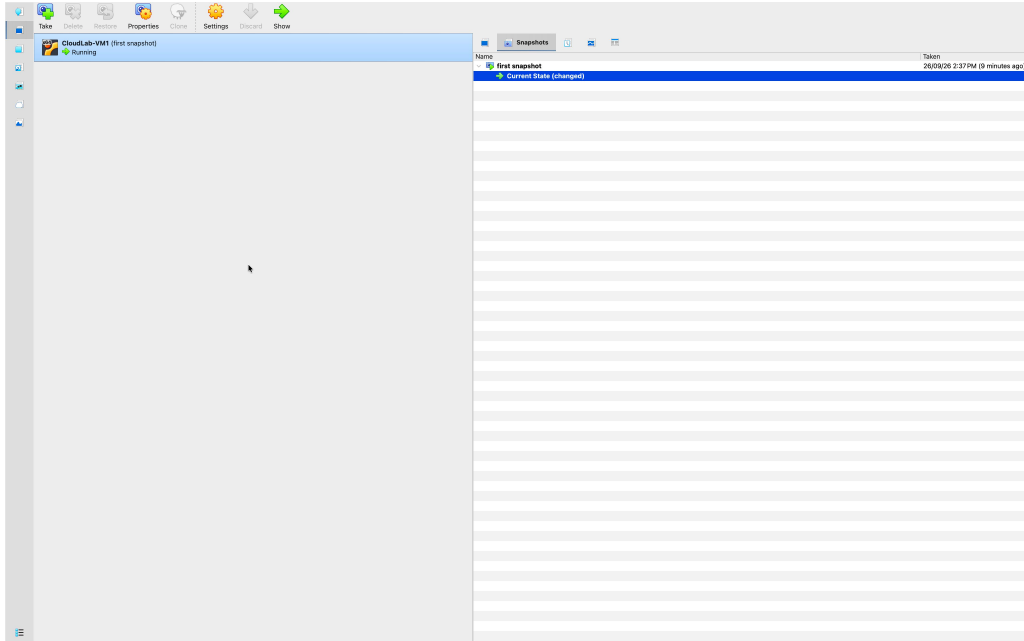


Figure 2: VirtualBox Snapshots tab showing `first snapshot` taken for CloudLab-VM1.

age, dynamically allocating IP addresses, and starting the instances without any human intervention.

Part 4 — Connecting to the Cloud (Option B: Case Study)

1. How does virtualization let the cloud provider improve hardware utilization compared to the company’s current setup?

Instead of the company having 20 separate physical servers sitting mostly idle at 15% utilization, virtualization allows the cloud provider to pool those resources. They can pack multiple different companies’ VMs onto fewer physical servers, running them at 80-90% utilization to maximize efficiency and save hardware costs.

2. What is “elasticity,” and how does the ability to clone/snapshot VMs support it?

Elasticity is the ability to automatically scale computing resources up or down based on current demand. Snapshots and cloning support this because new VMs can be cloned from a master snapshot and booted up instantly when traffic spikes, then deleted just as easily when traffic drops.

3. What risk does multi-tenancy introduce, and how does the hypervisor help mitigate it?

Multi-tenancy introduces security risks (one tenant potentially accessing another’s data) and performance risks (one tenant hogging the CPU, known as the “noisy neighbor” problem). The hypervisor mitigates this by creating strict, hardware-enforced boundaries—isolating CPU, memory, and network traffic so VMs cannot interact with or steal resources from each other.

Appendix: Lab Proof Screenshots

Note: The following screenshots demonstrate the successful creation, booting, snapshotting, and cloning of the virtual machine as required by the lab steps.

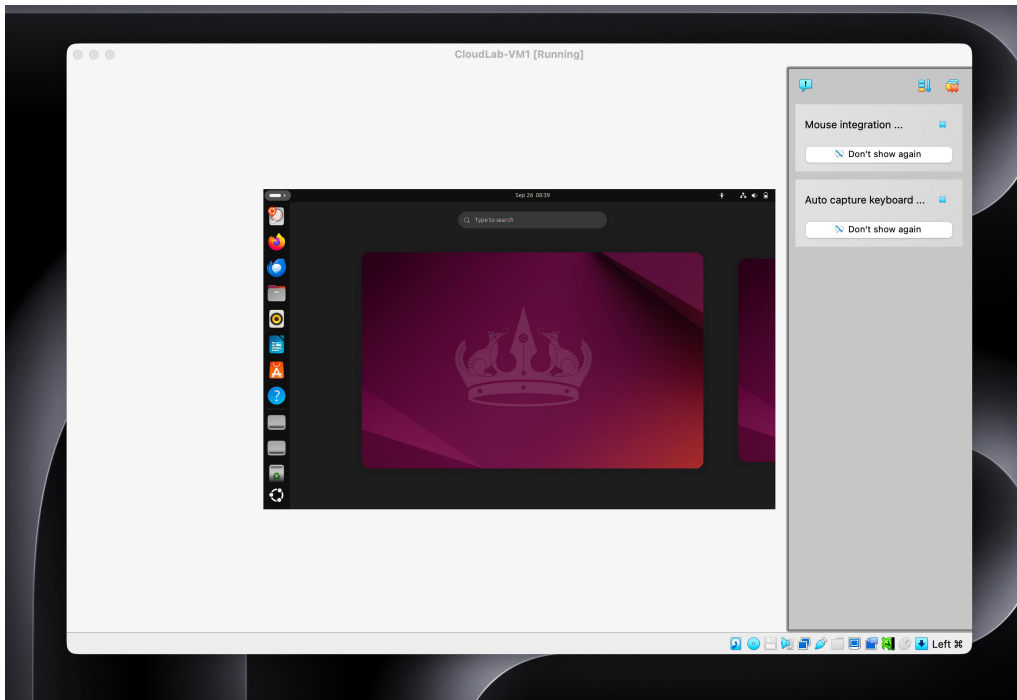


Figure 3: CloudLab-VM1 booted successfully to the Ubuntu 24.04 desktop environment.

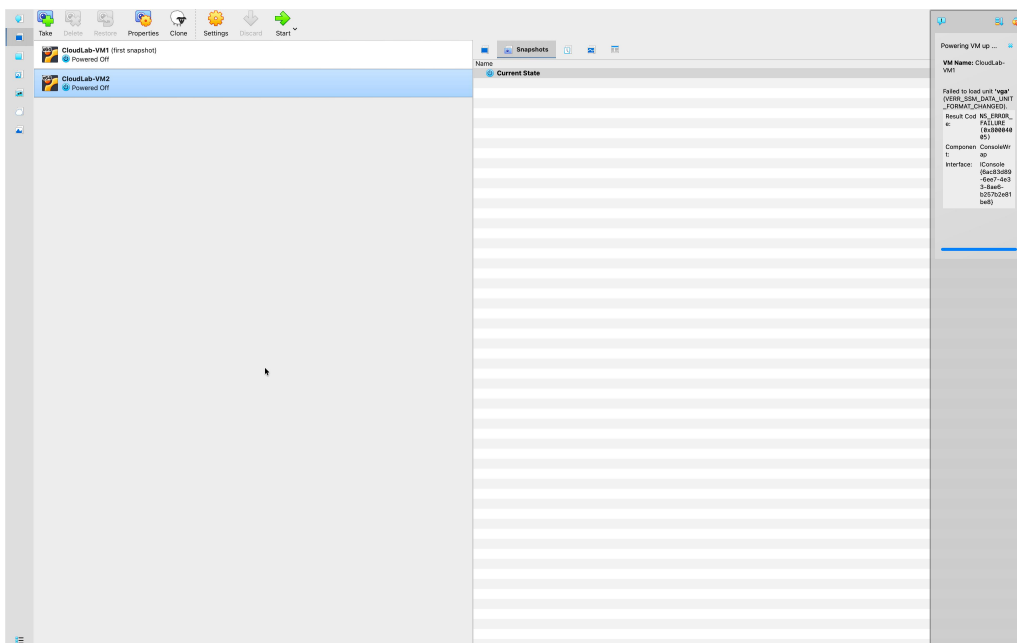


Figure 4: VirtualBox Manager showing both CloudLab-VM1 (original) and CloudLab-VM2 (full clone).